



Department of Computing

Project Part 2 – Data Preparation

Jose F. Barquero Jimenez
Brittany Isabell
Sharon Smith

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Instructor: Dr. Stephen Hendrix
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Table of Contents

- 1. DATA SOURCES 2
 - 1.1 Dataset 1: NBA Player Salaries 2024-25 3
 - 1.2 Dataset 2: Basketball Reference Advanced Stats 2024-25 4
 - 1.3 Dataset 3: NBA Per-Game Stats 2024 8
 - 1.4 Final, Unified Dataset 11
- 2. DATA PREPARATION..... 16
 - Part 1: Extraction..... 16
 - Part 2: Exploratory Data Analysis..... 16
 - Part 3: Transformation 17
 - Part 4: Loading..... 19
 - Part 5: KPI Calculations..... 19
- 3. ISSUES AND CHALLENGES 22

1. DATA SOURCES

The NBA Talent Acquisition Decision Support System (DSS) draws on three complementary datasets, all sourced from Kaggle and Basketball Reference, and covering the 2024-25 NBA season. Each dataset is provided in CSV format and is joinable to the others on the Player name field. Dataset 1 contains 517 unique players; Dataset 2 contains 569 unique players; and Dataset 3 contains 513 unique players. The final unified dataset contains 434 players with complete records across all three sources after incomplete records were removed to support full KPI analysis.

1.1 Dataset 1: NBA Player Salaries 2024-25

Source: Kaggle - <https://www.kaggle.com/datasets/ratin21/nba-player-salaries-2024-25>

Format: CSV

Records: 563 rows, 3 fields

Unique players: 517

What a single record represents: One player's contracted annual salary for the 2024-25 NBA season. This dataset is the primary financial input for all salary efficiency and cost-per-win KPI calculations. Players may appear multiple times if they have played for multiple teams during the season.

Table 1. *Dataset 1: NBA Player Salaries 2024-25, Field Descriptions*

Field Name	Data Type	Range / Valid Values	Description
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Player	String	Open-ended (player full name)	Full name of the NBA player. Serves as the primary join key between all three datasets.
Team	String	30 valid NBA team abbreviations (e.g., GSW, LAL, BOS)	Three-letter abbreviation of the team the player is contracted to for the 2024-25 season.
Salary	String (currency)	\$11,997 – \$55,761,216	Player's annual salary, formatted as a dollar string with commas. Must be cleaned (strip \$ and commas) before numeric analysis.

1.2 Dataset 2: Basketball Reference Advanced Stats 2024-25

Source: Basketball Reference-

https://www.basketballreference.com/leagues/NBA_2025_advanced.html

Format: CSV

Records: 736 rows, 30 fields

Unique players: 569 (736 rows due to mid-season trades creating multiple rows per player; use total rows for season-level analysis; league average row also included)

What a single record represents: One player's advanced efficiency statistics for the 2024-25 season. Players traded mid-season appear multiple times, once per team plus a total row, indicated by '2TM' (2 teams) or '3TM' (3 teams), for the full season. The total row should be used for season-level KPI analysis to avoid double-counting. This is the primary dataset for PER, Win Shares, VORP, and all percentage-based efficiency metrics.

Table 2. *Dataset 2: Basketball Reference Advanced Stats 2024-25, Field Descriptions*

Field Name	Data Type	Range / Valid Values	Description
Rk	Integer	1 – 736	Row rank. Display only; not analytically meaningful.
Player	String	Open-ended	Full player name. Primary join key. Players traded mid-season may appear multiple times — once per team plus a season total (TOT) row.
Age	Decimal	19 – 40	Player's age as of February 1 of the current season.
Team	String	30 NBA team abbreviations; 'TOT' for traded players	'TOT' indicates a season total row for players on multiple teams.
Pos	String	PG, SG, SF, PF, C	Primary position. Used as a filter for positional gap analysis.
G	Integer	1 – 82	Total games played. Used to filter players with insufficient sample size.
GS	Integer	0 – 82	Games started. Indicates starter vs. reserve role.
MP	Integer	2 – 3,036	Total minutes played. Used as a reliability filter for KPI comparisons.

PER	Decimal	-15.3 – 53.6 (league avg = 15.0)	Player Efficiency Rating. Composite per-minute performance metric. Primary KPI for player evaluation.
TS%	Decimal	0.000 – 1.500	True Shooting Percentage. Shooting efficiency accounting for 2-pointers, 3-pointers, and free throws. Values above 0.600 indicate elite efficiency.
3PAr	Decimal	0.000 – 1.000	Three-Point Attempt Rate. Proportion of field goal attempts that are three-pointers.
FTr	Decimal	0.000 – 2.000	Free Throw Attempt Rate. Ratio of free throw attempts to field goal attempts.
ORB%	Decimal	0.0 – 54.8	Offensive Rebound Percentage. Estimated percentage of available offensive rebounds grabbed.
DRB%	Decimal	0.0 – 55.8	Defensive Rebound Percentage. Key metric for team gap analysis in frontcourt roles.
TRB%	Decimal	0.0 – 30.4	Total Rebound Percentage. Combined rebounding rate used in positional need scoring.
AST%	Decimal	0.0 – 65.0	Assist Percentage. Estimated percentage of teammate field goals assisted. Primary playmaking metric.

STL%	Decimal	0.0 – 12.2	Steal Percentage. Estimated percentage of opponent possessions ending in a steal.
BLK%	Decimal	0.0 – 10.4	Block Percentage. Estimated percentage of opponent two-point attempts blocked. Key rim protection metric.
TOV%	Decimal	0.0 – 100.0	Turnover Percentage. Estimated turnovers per 100 plays. Lower is better.
USG%	Decimal	0.0 – 42.2	Usage Percentage. Estimated percentage of team plays used by the player. Contextualizes all other metrics.
OWS	Decimal	-2.3 – 12.7	Offensive Win Shares. Estimated wins contributed through offensive production.
DWS	Decimal	-0.1 – 4.8	Defensive Win Shares. Estimated wins contributed through defensive production.
WS	Decimal	-1.3 – 16.7	Win Shares. Total wins contributed (OWS + DWS). Primary KPI for salary efficiency modeling.
WS/48	Decimal	-0.55 – 0.60 (league avg $\approx$ 0.100)	Win Shares per 48 minutes. Volume-independent efficiency measure used in trajectory analysis.
OBPM	Decimal	-21.9 – 37.9	Offensive Box Plus/Minus. Estimated offensive points contributed per 100 possessions above league average.

DBPM	Decimal	-10.8 – 16.1	Defensive Box Plus/Minus. Estimated defensive points contributed per 100 possessions above league average.
BPM	Decimal	-32.7 – 44.1	Box Plus/Minus. Combined on-court impact per 100 possessions relative to league average.
VORP	Decimal	-1.6 – 9.8	Value Over Replacement Player. Points per 100 team possessions above a replacement-level player. Key KPI for acquisition value scoring.
Awards	String	Open-ended abbreviation string (e.g., MVP-7, AS, DPOY)	Season awards and All-Star selections. Optional contextual field.
Player-additional	String	Open-ended (Basketball Reference player ID)	Internal Basketball Reference player ID. Not used for joining to other datasets.

1.3 Dataset 3: NBA Per-Game Stats 2024

Source: Kaggle - <https://www.kaggle.com/datasets/columbiaave/nba-2024-2025-season-stats>

Format: CSV

Records: 528 rows, 31 fields

Unique players: 513

What a single record represents: One player's per-game counting statistics averaged across games played during the 2024-25 mid-season snapshot. Traded players appear multiple times —

once per team plus a total row, indicated by ‘2TM’ (2 teams), for the full season. The total row should be used to avoid double-counting. The maximum games played value of 35 confirms this is a partial season capture. All numeric fields are per-game averages except G (games played) and GS (games started). This dataset provides volume scoring and counting stats that complement Dataset 2's efficiency metrics.

Table 3. *Dataset 3: NBA Per-Game Stats 2024, Field Descriptions*

Field Name	Data Type	Range / Valid Values	Description
Rk	Decimal	1 – 528	Row rank. Display only; not analytically meaningful.
Player	String	Open-ended	Full player name. Primary join key.
Age	Decimal	19 – 40	Player's age as of February 1 of the season.
Team	String	30 NBA team abbreviations; 'TOT' for multi-team players	Team abbreviation.
Pos	String	PG, SG, SF, PF, C	Primary position. Used for positional filtering and gap analysis.
G	Decimal	1 – 35	Games played. Maximum of 35 indicates a mid-season snapshot, not full season.
GS	Decimal	0 – 35	Games started within the snapshot period.

MP	Decimal	1.0 – 39.3	Average minutes per game. Per-game average unlike Dataset 2, which stores total minutes.
FG	Decimal	0.0 – 12.7	Field goals scored per game.
FGA	Decimal	0.0 – 24.6	Field goal attempts per game.
FG%	Decimal	0.000 – 1.000 (5 nulls)	Field goal percentage.
3P	Decimal	0.0 – 4.5	Three-pointers scored per game.
3PA	Decimal	0.0 – 13.0	Three-point attempts per game.
3P%	Decimal	0.000 – 1.000 (36 nulls)	Three-point percentage. Null for players with no attempts.
2P	Decimal	0.0 – 12.5	Two-pointers made per game.
2PA	Decimal	0.0 – 20.3	Two-point attempts per game.
2P%	Decimal	0.000 – 1.000 (17 nulls)	Two-point field goal percentage.
eFG%	Decimal	0.000 – 1.500 (5 nulls)	Effective Field Goal Percentage. Adjusts for the added value of three-pointers.
FT	Decimal	0.0 – 7.8	Free throws made per game.
FTA	Decimal	0.0 – 11.8	Free throw attempts per game.
FT%	Decimal	0.000 – 1.000 (57 nulls)	Free throw percentage. Null for players with no attempts.
ORB	Decimal	0.0 – 3.9	Offensive rebounds per game.
DRB	Decimal	0.0 – 11.1	Defensive rebounds per game.

TRB	Decimal	0.0 – 14.0	Total rebounds per game.
AST	Decimal	0.0 – 11.9	Assists per game.
STL	Decimal	0.0 – 3.2	Steals per game.
BLK	Decimal	0.0 – 3.9	Blocks per game.
TOV	Decimal	0.0 – 4.8	Turnovers per game.
PF	Decimal	0.0 – 4.0	Personal fouls per game.
PTS	Decimal	0.0 – 32.4	Points per game. Primary volume scoring indicator.
Awards	Decimal / String	Largely null in this dataset	Award data field. Largely unpopulated; awards data is more complete in Dataset 2.

1.4 Final, Unified Dataset

Format: CSV

Records: 434 rows, 34 fields (38 after 4 additional variables were calculated in the unified database).

Unique players: 434

What a single record represents: One player's salary, their advanced efficiency statistics for the 2024-25 season, and their per-game counting statistics averaged across games played during the 2024-25 mid-season snapshot.

Table 4. *Dataset 4: Final, Unified Dataset*

Field Name	Data Type	Source Dataset	Description
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Player	String	DS2	Full player name, normalized to uppercase. Primary join key across all three datasets.
Age	Decimal	DS2	Player's age as of February 1 of the season.
Team	String	DS2	NBA team abbreviation. TOT for players traded mid-season.
Pos	String	DS2	Primary position: PG, SG, SF, PF, or C.
G	Integer	DS2	Total games played during the season.
GS	Integer	DS2	Total games started during the season.
MP	Integer	DS2	Total minutes played for the full season. Note: Dataset 3 stores MP as a per-game average — the Dataset 2 total is used as the authoritative source.
PER	Decimal	DS2	Player Efficiency Rating. Primary KPI for player evaluation. League average is 15.0.
TS%	Decimal	DS2	True Shooting Percentage. Shooting efficiency accounting for 2-pointers, 3-pointers, and free throws.
3PAr	Decimal	DS2	Three-Point Attempt Rate. Proportion of field goal attempts that are three-pointers.
FTTr	Decimal	DS2	Free Throw Attempt Rate. Ratio of free throw attempts to field goal attempts.
ORB%	Decimal	DS2	Offensive Rebound Percentage.
DRB%	Decimal	DS2	Defensive Rebound Percentage.

TRB%	Decimal	DS2	Total Rebound Percentage.
AST%	Decimal	DS2	Assist Percentage. Primary playmaking metric.
STL%	Decimal	DS2	Steal Percentage.
BLK%	Decimal	DS2	Block Percentage. Key rim protection metric.
TOV%	Decimal	DS2	Turnover Percentage. Lower is better.
USG%	Decimal	DS2	Usage Percentage. Contextualizes all other efficiency metrics.
OWS	Decimal	DS2	Offensive Win Shares. Wins contributed through offense.
DWS	Decimal	DS2	Defensive Win Shares. Wins contributed through defense.
WS	Decimal	DS2	Total Win Shares (OWS + DWS). Primary KPI for salary efficiency modeling.
WS/48	Decimal	DS2	Win Shares per 48 minutes. Volume-independent efficiency measure.
OBPM	Decimal	DS2	Offensive Box Plus/Minus.
DBPM	Decimal	DS2	Defensive Box Plus/Minus.
BPM	Decimal	DS2	Box Plus/Minus. Combined on-court impact per 100 possessions relative to league average.
VORP	Decimal	DS2	Value Over Replacement Player. Key KPI for acquisition value scoring.
Salary_USD	Integer	DS1	Annual salary in USD. Cleaned from original string format by removing

			dollar sign and commas. Stored alongside original Salary column for verification.
PTS	Decimal	DS3	Points per game. Primary volume scoring indicator.
TRB	Decimal	DS3	Total rebounds per game.
AST	Decimal	DS3	Assists per game.
STL	Decimal	DS3	Steals per game.
BLK	Decimal	DS3	Blocks per game.
TOV	Decimal	DS3	Turnovers per game.
Cost_per_WS	Decimal	Calculated	Annual salary divided by Win Shares. Measures salary expenditure per unit of win contribution.
Performance_Trajectory	String	Calculated	Classifies each player by career phase and WS/48 efficiency relative to their age peer group average.
PNS	Decimal	Calculated	Positional Need Score. A weighted composite of per-game statistics (PTS, TRB, AST, STL, BLK) with weights varying by position group.
Salary_Gap	Decimal	Calculated	Estimated difference between a player's actual salary and their market value, derived by scaling their Win Shares against the database maximum WS and maximum salary

2. DATA PREPARATION

All three datasets were joinable on the Player field (full player name string). Coverage across datasets was as follows: 468 players shared between Datasets 1 and 2; 438 between Datasets 1 and 3; and 491 between Datasets 2 and 3. There were 434 players present across all three datasets. Non-overlap was primarily caused by mid-season transactions and minor name formatting differences (accented characters, suffixes such as Jr. or III).

The Extract, Transform, Load (ETL) process was applied to produce a final, unified dataset for use during analysis of the Key Performance Indicators (KPI). A left-join strategy was used to maximize player coverage, with incomplete records removed at the end to ensure all 434 final players have complete data across all seven KPIs.

Part 1: Extraction

Each of the first three datasets outlined in the previous section were downloaded in CSV format from their specified external source websites. All three were imported into a Jupyter Notebook environment using Python.

Part 2: Exploratory Data Analysis

After extraction, several exploration steps were performed separately on each dataset to understand data quality and structure, and to better inform transformation needs. Exploration included review of data types, sample records, and data frame shape. Additionally, duplicates were investigated, and the number of unique players in each dataset was confirmed. Rows indicative of totals for multi-team players were also identified, along with rows displaying league average statistics rather than individual player statistics.

Part 3: Transformation

Each dataset was cleaned and transformed separately using the pandas library prior to completing any mergers. Before any cleaning steps were applied, a copy of each original DataFrame was created using the pandas `copy()` method so that all transformations were performed on the copy and the raw loaded data was preserved as a reference.

Dataset 1 with player salary information included duplicate player rows for multi-team players. Validation was performed to confirm that the salary amounts for these players were consistent across all records, and then the duplicate rows were dropped. A new `Salary_USD` column was created to store the cleaned integer value alongside the original `Salary` string column, preserving the original for verification rather than overwriting it. Player names were normalized by converting all names to uppercase and stripping white space.

Dataset 2 containing player advanced statistics required several cleaning steps. To begin, a row with league average statistics identified during exploration was dropped. Total rows for multi-team players (identified by '2TM' or '3TM' in the `Team` field) were standardized by replacing with the value 'TOT', for total. Filtering was then applied to the dataset to ensure that only 'TOT' rows for multi-team players were retained, along with all other single team players. The player rank field (`Rk`) was used to ensure maintenance of only one record per player during application of the filter. The `Rk`, `Player-additional`, and `Awards` columns were also dropped from Dataset 2 before the join. Dropping these before merging is critical because retaining them causes pandas to create duplicate columns with `_x` and `_y` suffixes wherever column names collide across datasets, resulting in over 60 columns of noise rather than a clean output. The `Rk` field was used as the player identifier during the traded player filtering step rather than player name string matching, because accented characters in some names such as Nikola Jokić and Dennis Schroder are stored differently across

the two datasets and can cause string-based matching to fail silently, dropping players from the result without any error message.

Lastly, player names were again standardized in this dataset using the same stripping and uppercase conversion applied to Dataset 1.

Transformation steps for Dataset 3 (per game stats) were very similar to those required for Dataset 2. The league average row was dropped, and again, replacement was used to standardize total rows for multi-team players. Filtering was applied through the same method to preserve only one record per player, keeping the 'TOT' row for multi-team players. The same player name normalization steps were also applied. Shooting percentage null values in the 3P%, 2P%, and FT% columns were intentionally left as NaN and not filled. These nulls exist because some players have zero attempts in that category, a center who never attempts three-pointers will correctly have 3PA equal to zero and 3P% equal to NaN. Filling these values with zero or a column average would misrepresent those players' shooting profiles.

Finally, the cleaned datasets were merged through two steps. To provide the broadest range of coverage, the cleaned datasets were merged using a left-join strategy anchored on Dataset 2, which has the largest unique player count at 569 and contains all primary KPI columns including PER, Win Shares, BPM, and VORP. Dataset 2 was chosen as the anchor because anchoring on the dataset with the most analytical value ensures no KPI-relevant records are lost as a starting point. Dataset 1 was joined first, bringing only the Salary_USD column to avoid duplicate column names. Dataset 3 was joined second, bringing only PTS, TRB, AST, STL, BLK, and TOV. Selecting only the specifically needed columns from each supporting dataset before merging prevented duplicate column name collisions. A left join was used rather than an inner join to preserve all Dataset 2 players even where salary or per-game data was unavailable.

Part 4: Loading

The clean, unified dataset was exported from Jupyter Notebook by writing it to a CSV file. As a final step, four additional variables were added directly into the CSV to provide calculations for KPIs that were not already included in the dataset.

Players with null values in the Salary_USD or PTS fields were removed from the final dataset to ensure complete records for all seven KPI calculations required by the DSS. This reduced the player count from 569 to 434. The final dataset contains one unified record for 434 unique players with complete data across all fields, and includes all necessary variables for analysis of the seven KPIs established during the DSS feasibility study. A sample of the first 100 records is provided as a separate attachment, NBA_DSS_UnifiedDf_Sample.csv

Part 5: KPI Calculations

Upon completion of the ETL process, the unified dataset contained 34 fields (columns A through AH). The CSV file was then exported from Python and opened in Microsoft Excel for finalization. Four additional derived columns were added directly in Excel to ensure all seven KPIs identified in the feasibility study are present and computable. For KPI 5 (Performance Trajectory), a dedicated reference table was created on a second worksheet titled Reference table, which dynamically computes the average WS/48 for each age group from the live dataset. Table 5 summarizes the source and status of all seven KPIs in the final database.

Cost per win contribution was entered in cell AI2 and filled down through row 435 using the following formula: $=IF(V2>0, AB2/V2, "N/A")$, where $V = WS$; $AB = Salary_USD$; *IF guard returns "N/A" for the players with $WS \leq 0$.*

A reference table worksheet to store dynamic WS/48 peer averages by age group was used for Performance_Trajectory KPI. The following formulas was used on the reference table:

1. Rising (age ≤ 25):

=AVERAGEIF(NBA_DSS_UnifiedDf!B2:B435,"<=25",NBA_DSS_UnifiedDf!W2:W435).

2. Peak (age 26–29):

=AVERAGEIFS(NBA_DSS_UnifiedDf!W2:W435,NBA_DSS_UnifiedDf!B2:B435,">=26",NBA_DSS_UnifiedDf!B2:B435,"<=29")

3. Veteran (age ≥ 30):

=AVERAGEIF(NBA_DSS_UnifiedDf!B2:B435,">=30",NBA_DSS_UnifiedDf!W2:W435)

Then the following formula was used to calculate the performance trajectory: =IF(B2<=25, IF(W2>='Reference table'!\$B\$2,"Rising - Above Avg","Rising - Below Avg"), IF(B2<=29, IF(W2>='Reference table'!\$B\$3,"Peak - Above Avg","Peak - Below Avg"), IF(W2>='Reference table'!\$B\$4,"Veteran - Above Avg","Veteran - Below Avg"))); where B = Age; W = WS/48; \$B\$2:\$B\$4 = locked peer average cells on the Reference table sheet.

Table 5. KPI Sources and Descriptions

KPI	Column	Source	Type	Status
KPI 1 — Player Efficiency Rating (PER)	H	Basketball Reference (Dataset 2)	Pre-existing	Ready
KPI 2 — Win Shares (WS) & VORP	V, AA	Basketball Reference (Dataset 2)	Pre-existing	Ready
KPI 3 — Cost per Win Contribution	AI	Derived: Salary ÷ WS	Computed	Computed

KPI 4 — Box Plus/Minus (BPM)	Z	Basketball Reference (Dataset 2)	Pre-existing	Ready
KPI 5 — Performance Trajectory	AJ	Derived: Age + WS/48	Computed	Computed
KPI 6 — Positional Need Score (PNS)	AK	Derived: Weighted per-game stats	Computed	Computed
KPI 7 — Salary vs. Market Value Gap	AL	Derived: Salary – WS-scaled value	Computed	Computed

Positional need score (KPI 6) and Salary vs Market value Gap (KPI 7) were calculate in cells AK2 and AL2 respectively, using the following formulas:

1. KPI6: =IF(OR(D2="PG",D2="SG"), (AE2*0.35)+(AC2*0.35)+(AF2*0.20)+(AD2*0.10), IF(D2="SF", (AC2*0.30)+(AD2*0.25)+(AE2*0.20)+(AF2*0.15)+(AG2*0.10), (AD2*0.35)+(AG2*0.30)+(AC2*0.20)+(AF2*0.15))); where D = Pos; AC = PTS; AD = TRB; AE = AST; AF = STL; AG = BLK.
2. KPI 7: =IF(V2>0, AB2-((V2/MAX(\$V\$2:\$V\$435))*MAX(\$AB\$2:\$AB\$435)), "N/A"); where V = WS; AB = Salary_USD; MAX ranges are locked so the ceiling values do not shift when filling down; Positive = overpaid; Negative = undervalued; Returns "N/A" for players with $WS \leq 0$.

3. ISSUES AND CHALLENGES

An initial challenge was faced during the data importation process into Jupyter Notebook. Dataset 2 importation failed due to a utf-8 encoding error. It is possible this occurred due to special characters found in player's names (e.g., symbols, accents), but the problem was easily resolved by adding information to the code to specify encoding.

A second challenge was the discovery of a League Average summary row in both Dataset 2 and Dataset 3. Basketball Reference automatically appends this row to every statistical table it exports. Because it is not a player record, it would corrupt any league-wide average or ranking calculation if retained. It was removed by filtering for rows where the Player field did not equal "League Average" before any join was performed.

A third challenge involved column name collisions during the merge step. Datasets 1 and 3 share several column names with Dataset 2, including Age, Team, Pos, G, GS, and MP. When all columns from every dataset are joined without selection, pandas automatically appends `_x` and `_y` suffixes to distinguish them, producing a messy output with duplicate columns. This was resolved by selecting only the specifically needed columns from Datasets 1 and 3 before merging.

A fourth challenge was a units mismatch in the MP column. Dataset 2 stores MP as total season minutes with values up to 3,036. Dataset 3 stores MP as a per-game average with values up to 39.3. Both columns share the same name but measure completely different quantities. Dataset 2's total minutes were used as the authoritative source in the unified dataset for three reasons: Dataset 2 covers the full season while Dataset 3 is a mid-season snapshot, the primary KPI calculations require total minutes rather than per-game averages, and Dataset 2 serves as the anchor for the entire join so its values form the baseline for all shared columns. Dataset 3's MP field was excluded accordingly.

A fifth challenge involved the transparency of the data filtering process. An initial pipeline approach using inner joins produced a final dataset of 434 players, but did so by silently dropping 135 players without any record of which players were removed or why. Switching to a left join retained all 569 players initially, making the null values visible and allowing an informed decision

about which records to exclude. Players missing Salary_USD or PTS values were then explicitly removed in a documented step, also producing 434 players. Although the final count is the same, the left join approach provides a transparent and auditable record of every exclusion decision, which is essential for a decision support system where data integrity must be verifiable.